The Phase-Field Method: Simulation of Alloy Dendritic Solidification during Recalescence

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An overview of the phase-field method for modeling solidification is given and results for nonisothermal alloy dendritic growth are presented. By defining a "phase-field" variable and a corresponding governing equation to describe the state (solid or liquid) in a material as a function of position and time, the diffusion equations for heat and solute can be solved without tracking the liquid-solid interface. The interfacial regions between liquid and solid involve smooth, but highly localized variations of the phase-field variable and the composition. Simple finite-difference techniques on a uniform mesh can be used to treat the evolution of complex growth patterns. However, large-scale computations are required. The method has been applied to a variety of problems, including thermally driven dendritic growth in pure materials, solute-driven isothermal dendritic growth in alloys, eutectic growth (all at high supercoolings or supersaturations), solute trapping at high velocity, and coarsening of liquid-solid mixtures. To include thermal effects in solute-driven dendritic growth in alloys, a simplified approach is presented here that neglects the spatial variation of temperature in the computational domain but provides for changes with time and thus includes recalescence. Growth morphologies and solute patterns in the liquid and solid obtained for several values of an imposed heat flux are compared to results for isothermal growth.

I. INTRODUCTION

IN a recent series of articles, a new method for computing solidification morphology in binary alloys has been developed for single-phase growth[1–5] and coarsening,[6] and for eutectic growth.[7–10] These articles extend research on pure materials[11–17] that employ a phase-field variable, ϕ, which is a function of position and time, to describe whether the material is liquid or solid. An extra equation is developed to govern the behavior of this new variable in addition to equations for heat and solute transport. Interfaces between liquid and solid are described by smooth but highly localized changes of this variable between values that represent liquid and solid (often 0 and 1, respectively). Realistic simulations of dendritic growth in pure materials have been obtained[18–23] that naturally include the selection of tip radius and growth rate for prescribed values of the supercooling far from the growing dendrite.

For a binary alloy, Warren and Boettinger[4] have computed growth shapes and solute microsegregation patterns for two-dimensional dendrites growing into a highly supersaturated liquid using an isothermal model. The isothermal approximation necessarily ignores the release of latent heat. The present article gives an overview of the phase-field method and also gives new results for alloy dendritic growth that include a simplified model of the latent heat release. An overall heat balance in the computational domain that is subject to external cooling is used to govern the temperature, which is assumed to vary with time but not with position. Thus, recalescence behavior is obtained.

While the essential ingredients of phase-field methods have existed for many years, recent progress is directly related to the increased availability of computational resources. This increase has also lead to renewed attempts to perform solidification calculations using more traditional sharp interface methods as well. These methods require a solution to the difficult moving boundary problem and the application of boundary conditions at the interface. Techniques that treat the moving boundary problem with Green's functions,[24] front tracking,[25] or adaptive finite elements[26] have produced simulations of dendritic growth for pure materials that show much of the same detail obtained with the phase-field method. The phase-field method, while computationally intense, requires only the application of finite-difference techniques to the relevant equations. The location of the interface is obtained from the solution for the phase-field variable as positions where $\phi = 1/2$ for example. The phase-field method also permits coalescence of two solid regions that come into close proximity and easily treats topology changes. Using simple finite-difference algorithms with current computational resources, dendritic growth is most easily simulated at high supercoolings (supersaturations) because computations then require small domains and short time intervals.

II. THE PHASE-FIELD METHOD FOR PURE MATERIALS

A. *Temperature*

One can begin a discussion of the phase-field approach[14] with a description of the enthalpy method for solidification modeling.[27,28] In the simplest form of this method for a pure material, the temperature, T, is determined as weak solutions (in an integral sense, see Reference 29) of the energy equation

$$\frac{\partial H}{\partial t} = k\nabla^2 T \qquad [1]$$

using an enthalpy, H, given by

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$$H = H_0 + C_pT + L \quad \text{if } T > T_M$$
$$H = H_0 + C_pT \quad \text{if } T < T_M \quad [2]$$

where k is the thermal conductivity, H_0 is a constant, C_p is the heat capacity per unit volume that in general depends on temperature, L is the latent heat per unit volume, and T_M is the melting point. This method gives the position of the liquid-solid interface (the location of the T_M isotherm) without applying the flux continuity condition at the interface; however, the interface is always at T_M and no supercooling of the liquid or the interface (due to curvature and/or growth kinetics) is possible.

By introducing into the description of the enthalpy a variable, ϕ, that identifies the physical state (liquid or solid), that is independent of temperature, one obtains

$$H = H_0 + C_pT - L\phi \quad [3]$$

yielding an equation for thermal diffusion containing a source term given by

$$C_p \frac{\partial T}{\partial t} - L \frac{\partial \phi}{\partial t} = k\nabla^2 T \quad [4]$$

Clearly the latent heat evolution occurs where ϕ is changing with time. An identical equation is used in macroscopic modeling of heat flow during solidification. In such cases, ϕ would represent an average value, defined over some appropriate distance, of the fraction solid in a mushy region. In the present case, we consider ϕ to be a variable that varies continuously from 0 to 1 across a single liquid-solid interface. With the addition of an equation to describe the evolution of ϕ, as described subsequently, the phase-field method can treat supercooled liquids (and also superheated solids) as well as interfaces not constrained to be at T_M. More complete phase-field formulations of the energy equation and the enthalpy that include temperature and phase-dependent materials properties are possible.[30]

B. *The Phase Field*

Figure 1, adapted from Mikeev and Chernov,[31] shows one possible physical interpretation of the phase-field variable. An extremely simplistic view of the interfacial region and its motion during solidification is depicted by a damped wave that represents the atomic density as a function of distance between a crystalline material and a liquid of the same average densities. On the left, the atomic density is localized to discrete atomic planes corresponding to solid. As the liquid is approached, the density has the same average value but becomes less localized, as indicated by the reduced amplitude of the wave. Finally, the density achieves a constant value in the liquid, indicating the absence of localization. The amplitude of the wave might be interpreted as the phase field, ϕ. In the lower portion of Figure 1, the atomic motions that might accompany the movement of the interfacial region during a solidification process are shown. Another interpretation for ϕ can be obtained by considering the ordering of approximately equal concentrations of atoms and vacancies on a lattice. Here, a random mixture of atoms and vacancies would be considered the liquid.

Little is known experimentally about the structure of the interface between liquid and solid phases. The current

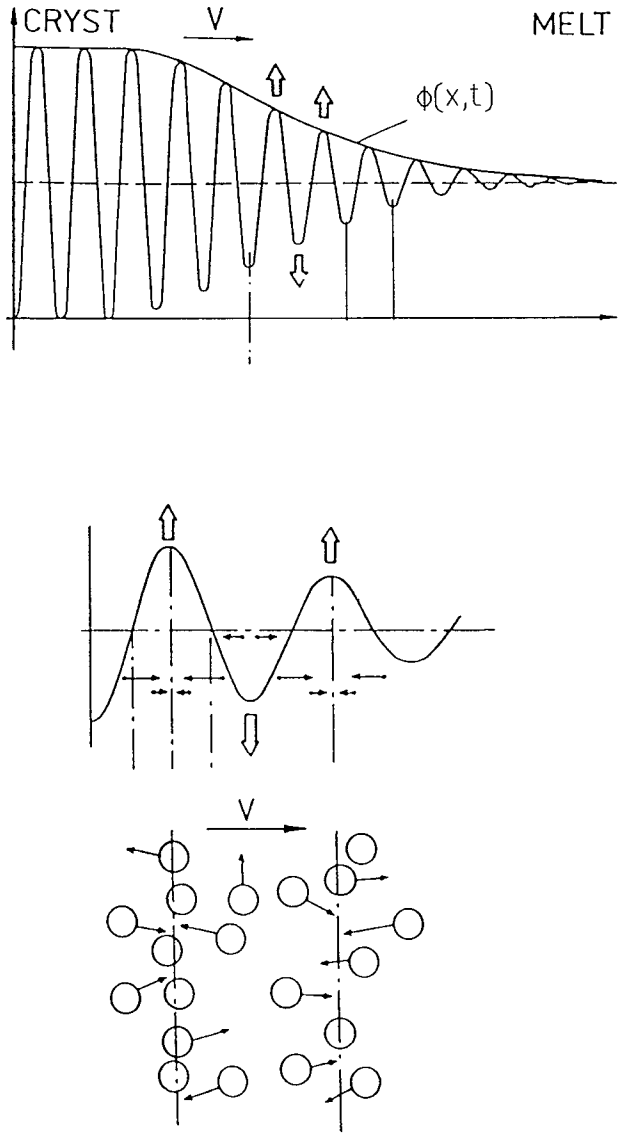


Fig. 1—Schematic representation of a possible physical interpretation of the phase-field variable (adapted from Meekev and Chernov[31]).

phase-field method uses a simple scalar-order parameter to describe the energy associated with the interfacial layer. Physically, more complex descriptions of the change in order across a liquid-solid interface are necessary. Indeed, density functional theory[32] considers a variety of approximations to treat the energy associated with the interfacial region. Even with the use of a single scalar-order parameter, the choice of a value for the interface thickness is not clear. Using Lennard–Jones potentials, molecular dynamics simulations of the transition in atomic positions across an interface have suggested that the interface width is several atomic dimensions.[33] As will be shown, calculations using the phase-field equations with an interface thickness of this order with present computational resources are only possible for one-dimensional problems or for very small two-dimensional regions. To simulate a full dendritic shape with side branches, thicker interfacial regions must be employed. Thus, it may be better at present to consider the phase-field parameter as a mathematical tool to allow easier calculations of solidification patterns that are useful for qualitative insight. It might be noted that the traditional "sharp inter-

face'' model is also only an approximation of physical reality, which is used to formulate the classical moving boundary problem.

C. *Evolution Equation for the Phase Field*

The evolution equation for ϕ is obtained using variational calculus for equilibrium and using gradient thermodynamics for kinetics. Although an entropy formulation is appropriate for nonisothermal systems,[30] a simpler isothermal formulation is described here for illustration. Equation [4] can then be appended to the governing equation for ϕ, giving essentially equivalent results to the entropy formulation.

Because the molar volume is assumed to be constant in the phase-field method, use of a Helmholtz energy is appropriate. We consider such an energy per unit volume, $f(\phi,T)$, such that $f(0,T)$ and $f(1,T)$ are the classical free energies of the liquid and solid, respectively, which are equal at the equilibrium melting point, T_M. Near the melting point their difference is given by

$$f(1,T) - f(0,T) = \frac{L(T - T_M)}{T_M} \qquad [5]$$

For values of ϕ not equal to zero or unity, the increased energy is assumed to be given by a ''double well'' potential, which at $T = T_M$, is given by

$$f(\phi,T_M) = \frac{W}{4}\phi^2(1 - \phi)^2 \qquad [6]$$

With this expression, the barrier height of the double well potential, $f(1/2,T_M) - f(0 \text{ or } 1,T_M)$, is $W/64$. The temperature dependence of the Helmholtz energy is taken in such a manner that the minima remain at $\phi = 0$ and 1 and the barrier never disappears. One choice is

$$f(\phi,T) = W\int_0^{\phi} [p(p - 1)\,(p - \frac{1}{2} - \beta(T)]dp \qquad [7]$$

which is the same as

$$f(\phi,T) = \frac{W}{4}\phi^2\left\{\phi^2 - [2 + \frac{4}{3}\beta(T)]\phi + [1 + 2\beta(T)]\right\} \qquad [8]$$

where β is given by

$$\beta(T) = \frac{6}{W}\frac{L(T - T_M)}{T_M} \qquad [9]$$

in order to guarantee that Eq. [5] is satisfied. The temperature is constrained to keep $\beta(T)$ between $-1/2$ and $+1/2$ to prevent the barrier from disappearing. Thus, we avoid phase transitions near critical points.

The governing equation for the phase field, ϕ, is obtained by considering a free-energy functional, F, constructed over the volume, V, of the entire system that includes not only the energy $f(\phi,T)$ but also an energy cost associated with gradients, $\nabla\phi$, given by

$$F[\phi,T] = \int_V [f(\phi,T) + \frac{\varepsilon^2}{2}|\nabla\phi|^2]dV \qquad [10]$$

The parameter ε is a gradient energy coefficient that is related to the interface energy and interface thickness. While more complex descriptions of the functional F are possible, this is the simplest form that will permit the existence of interfacial regions between phases.

For global equilibrium the temperature must be a constant, and F must be minimized with respect to the shape of the spatial profile of ϕ. The calculus of variations yields the Euler equation

$$\varepsilon^2\nabla^2\phi - \frac{\partial f}{\partial\phi} = 0 \qquad [11]$$

Trivial but important solutions to this equilibrium equation are obtained if ϕ is constant with values given by $\partial f/\partial\phi = 0$. By the construction of our form of $f(\phi,T)$, this situation corresponds to $\phi = 0$ (a single-phase liquid) or $\phi = 1$ (a single-phase solid). The equilibrium equation permits these states to exist at any temperature including those for a metastable supercooled liquid or a superheated solid.

A nontrivial one-dimensional (planar interface) solution to the equilibrium equation exists, however, only at $T = T_M$ and yields a transition zone between liquid ($\phi = 1$) and solid ($\phi = 0$), where ϕ varies as

$$\phi(x) = \frac{1}{2}[1 - \tanh(\frac{x}{2\delta})] \qquad [12]$$

Here, δ is a measure of the interface thickness and is given by

$$\delta = \frac{\varepsilon\sqrt{2}}{\sqrt{W}} \qquad [13]$$

The value of the interface thickness is a balance between two opposing effects. The interface tends to be sharp in order to minimize the volume of material where ϕ is between 0 and 1 and $f(\phi,T)$ is large (as described by W). The interface tends to be diffuse to reduce the energy associated with the gradient of ϕ (as described by ε). We note that the phase-field variable goes from 0.1 to 0.9 over a distance of 4δ.

Figure 2 shows schematically the variation of ϕ with distance as well as the integrand of Eq. [10], $f(\phi,T_M) + 1/2\,\varepsilon^2(\nabla\phi)^2$, across an interfacial region. The hatched region corresponds to the surface free energy and is given by Wheeler et al.[1] and others as

$$\sigma = \frac{\varepsilon\sqrt{W}}{6\sqrt{2}} \qquad [14]$$

Equations [13] and [14] can be used to calculate the values of the parameters ε and W to match with selected values for σ and δ. These equilibrium solutions, while physically obvious, are a requirement for any model of solidification.

For moving interfaces we make the simplest postulate that guarantees that global equilibrium will be obtained after long times (Eq. [11]) and that F is a decreasing function of time; *viz.*,

$$\frac{\partial\phi}{\partial t} = -M_\phi(\frac{\partial f}{\partial\phi} - \varepsilon^2\nabla^2\phi) \qquad [15]$$

where M_ϕ is a positive mobility constant. For an isothermal system at temperature T, Eq. [12] remains a solution to the

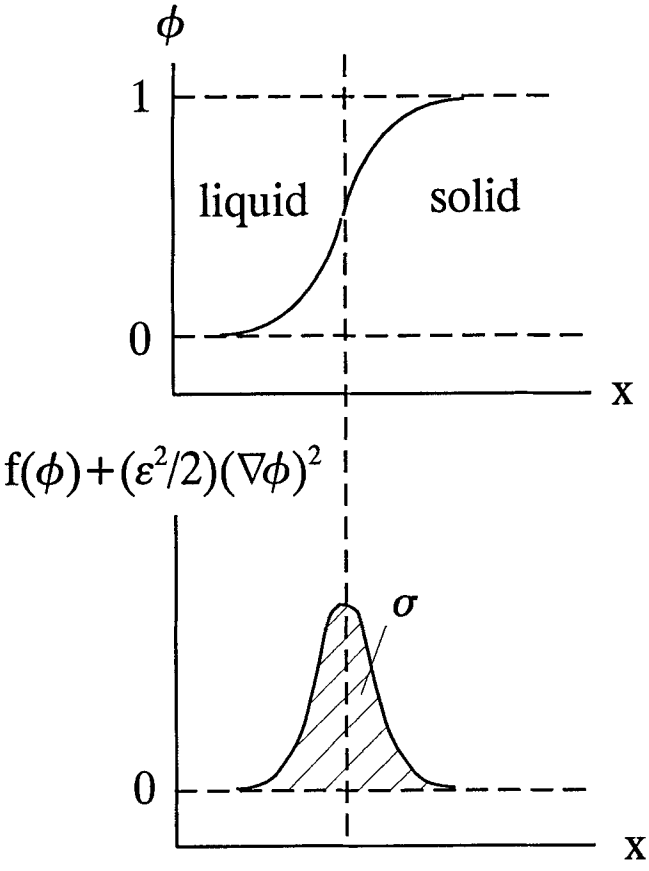


Fig. 2—Schematic variation of the phase-field parameter, ϕ, and the energy/unit volume, $f(\phi) + (1/2)\varepsilon^2 (\nabla\phi)^2$, with distance, x, across a stationary flat liquid-solid interface at the melting point in a pure material. The hatched region is the excess energy/unit area of interface, i.e., the liquid-solid interface energy.

dynamic equation (Eq. [15]) in a coordinate frame moving at constant velocity V.[34] However, to obtain this solution, V must be related to the temperature by

$$V = \mu(T_M - T) \qquad [16]$$

where

$$\mu = \frac{6M_\phi \varepsilon L\sqrt{2/W}}{T_M} \qquad [17]$$

This selection of one interface velocity for each (interface) temperature is consistent with the classical approach to linear interface attachment kinetics whose coefficient is μ. Equation [17] can be used to determine a value of M_ϕ from a knowledge (or more likely an estimate) of μ.

Equation [15] is known as the Cahn–Allen equation, which was developed for the motion of antiphase boundaries in ordered alloys.[35] In that case, the barrier height of $f(\phi,T)$ was chosen to vanish at a critical point consistent with the second-order transition treated. For the present application, as stated previously, we require that the barrier not vanish.

It is important to note that the limit $\delta \to 0$ can be taken while holding σ and μ fixed, a limit that should approach the classical sharp interface formulation of the solidification problem. Indeed, formal asymptotic analysis,[15,36] performed for nonplanar interfaces, recovers the local conditions

$$T_L = T_S = T_M - \frac{T_M \sigma}{L} K - \frac{V_n}{\mu} \qquad [18]$$

and

$$k(n \cdot \nabla T)\Big|_S^L = -LV_n \qquad [19]$$

where T_L and T_S are the temperatures of the liquid and solid at the interface, K is the mean curvature, and n is the normal to the interface, and V_n is the normal velocity.

The choice for the form of the function $f(\phi,T)$ is quite arbitrary. Other choices for the double-well are often employed in the literature, but one-dimensional analytical solutions of the governing equations have not been found. However, asymptotic analysis as the thickness goes to zero recovers the same local interface conditions, Eqs. [18] and [19]. The relations given by Eqs. [13], [14], and [17] are then altered by numerical factors related to the choice of the measure of the double-well height.

Numerical solutions of these (or similar) equations employing very small values of δ (approaching atomic dimensions) are possible with current computational resources in one dimension. Wheeler et al.[21] have obtained solutions for the growth of a spherical crystal into a supercooled melt that are in excellent agreement with the sharp interface Green's function calculations of Schaefer and Glicksman[37] that include the Gibbs–Thomson effect and interface kinetic undercooling.

D. *Other Simulation Issues*

To obtain "realistic" dendrites using the phase-field equations, three other issues must be addressed, anisotropy, noise, and computation times. The most widely used method to include anisotropy for two-dimensional calculations is to assume that ε in Eq. [10] depends on an angle θ.[20] Here, θ is the orientation of the interface and is computed through the expression

$$\tan\theta = \left(\frac{\partial\phi}{\partial y}\right)\Big/\left(\frac{\partial\phi}{\partial x}\right) \qquad [20]$$

where x and y define the coordinate system. This change requires a recomputation of the variational derivative in Eqs. [11] and [15] to produce a more complex field equation for ϕ. For numerical computations of dendrites, fourfold[20–23] and sixfold[22] variations of ε with θ have been employed. Using this method of including anisotropy, formal asymptotics, as the interface thickness is taken to zero, yield the same form of the anisotropic Gibbs–Thompson equation employed for sharp interface theories.[36]

Calculations for growth into an undercooled melt starting from a small solid ($\phi = 1$) seed with appropriate levels of anisotropy typically exhibit needlelike growth with parabolic growth forms in the directions with the smallest values of σ. Calculations performed using a coarse finite-difference mesh will usually exhibit side branching typical of real dendrites because discretization errors introduce noise into the calculation. (A coarse square mesh can also induce a synthetic four-fold symmetry.) As the computational mesh is refined, however, side branches necessarily disappear. Therefore, noise has been introduced at

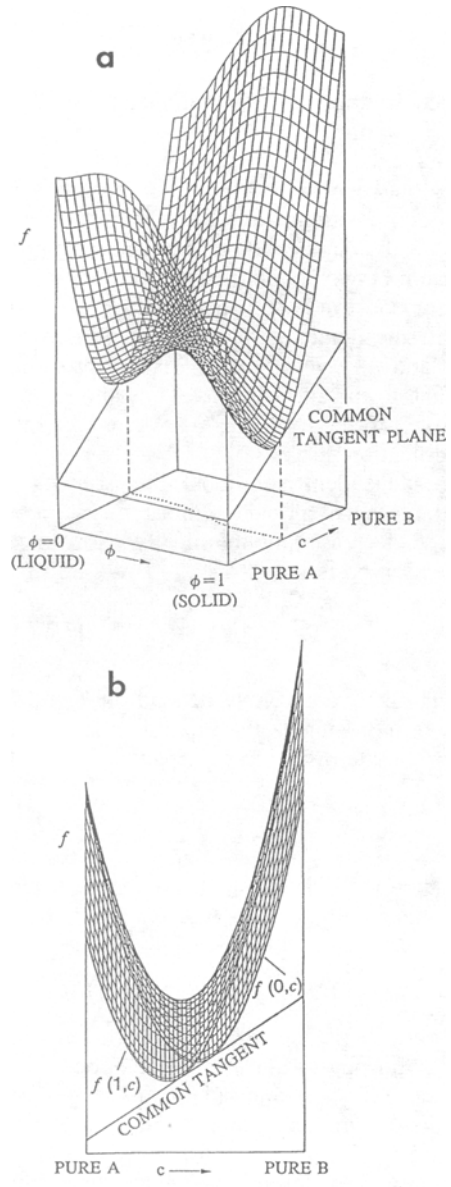


Fig. 3—(*a*) The alloy free-energy function $f(\phi,T,c)$ for a temperature, T, between the pure component melting points ($T_M^B < T < T_M^A$), showing the tangent plane that determines the compositions of liquid and solid far from a stationary flat interface. The locus of ϕ and c through the thin interfacial region is shown dotted. (*b*) Projection of (*a*) along the ϕ axis showing the common tangent line. The free-energy surface is only shown for $0 < \phi < 1$.

controlled levels to induce side branching in most simulations. This is typically implemented using random fluctuations of a source term added to the phase-field equation that is localized to regions where ϕ is between 0 and 1; i.e., in the interfacial region.[20] More physical descriptions of thermal noise using a Langevin approach have been used,[8] and this permits the inclusion of nucleation processes into simulations using the phase-field method. A thorough investigation of the interplay between interface thickness and noise amplitude should be conducted.

Computational issues are focused on the value of the interface thickness and the finite-difference mesh spacing. For stable calculations, typically the largest possible mesh spacing is taken to be equal to δ, giving approximately four mesh points from $0.1 < \phi < 0.9$. Because we cannot take δ to be equal to atomic dimensions without incurring excessive computation times for domains large enough for a dendrite with side branches (>>10 hours on CRAY-YMP*), we must presently accept more practical restrictions on δ. Certainly δ must be smaller than the tip radius of the dendrite. However, Braun et al.,[38] investigating growth in hypercooled melts, have shown that δ must be less than the capillary length, $\sigma T_M C_p/L^2$, (which is of the order of atomic dimensions) to ensure agreement with planar interface stability theory. For realistic choices of the interface energy and kinetic coefficient for metals (corresponding to collision limited growth), calculations of dendrites in pure materials are presently limited to an interface thickness in the range of 10 to 30 nm and to calculations where the supercooling is no smaller than $\sim 0.7L/C_p$ for CRAY-YMP runs on the order of 2 CPU hours at 200 MFLOPS.

*CRAY-YMP is a trademark of Cray Research, Inc., Minneapolis, MN.

With the choice of a relatively large value for δ, the tip radii (and velocities) obtained from phase-field calculations of dendrites will depend on δ and are larger (smaller) than would be obtained by experiment or from free-dendrite models that use the marginal stability criterion for tip radius selection.[39] By performing calculations in a computational domain that only treats the tip region, smaller values of δ can be used and one can achieve tip radii and velocities that are independent of δ.[40,41]

III. THE PHASE-FIELD MODEL FOR BINARY ALLOYS

To treat a binary alloy, an additional equation governing the concentration is required. This equation should reduce to the standard solute diffusion equation in the liquid and solid phases but is likely to have a different form through the interfacial region. Such an equation has been found by employing a Helmholtz energy that depends on c in addition to T and ϕ and using the same variational method.[1–5] The simplest form for f, corresponding to ideal liquid and solid solutions, is given by

$$f(\phi,c,T) = (1 - c)f_A(\phi,T) + cf_B(\phi,T) + R'T[(1 - c)\ln(1 - c) + c\ln c] \quad [21]$$

where R' is the universal gas constant per unit volume (molar volume is assumed constant), and T is the temperature. The functions f_A and f_B are the free energies of pure A and pure B components and may be given as previously (Eqs. [8] and [9]) with different melting points, T_M^A and T_M^B, and latent heats, L_A and L_B. The shape of the alloy free energy $f(\phi,c,T)$ as given by Eq. [21] at a fixed temperature between the pure component melting points is shown in Figure 3.

The free-energy functional including concentration is then given by

$$F[\varphi,c,T] = \int_V \left[f(\varphi,c,T) + \frac{\varepsilon^2}{2}|\nabla\varphi|^2\right]dV \quad [22]$$

Using the same approach as before, except employing the constraint that the total amount of solute in the volume V is a constant, a necessary condition on the concentration for global equilibrium is

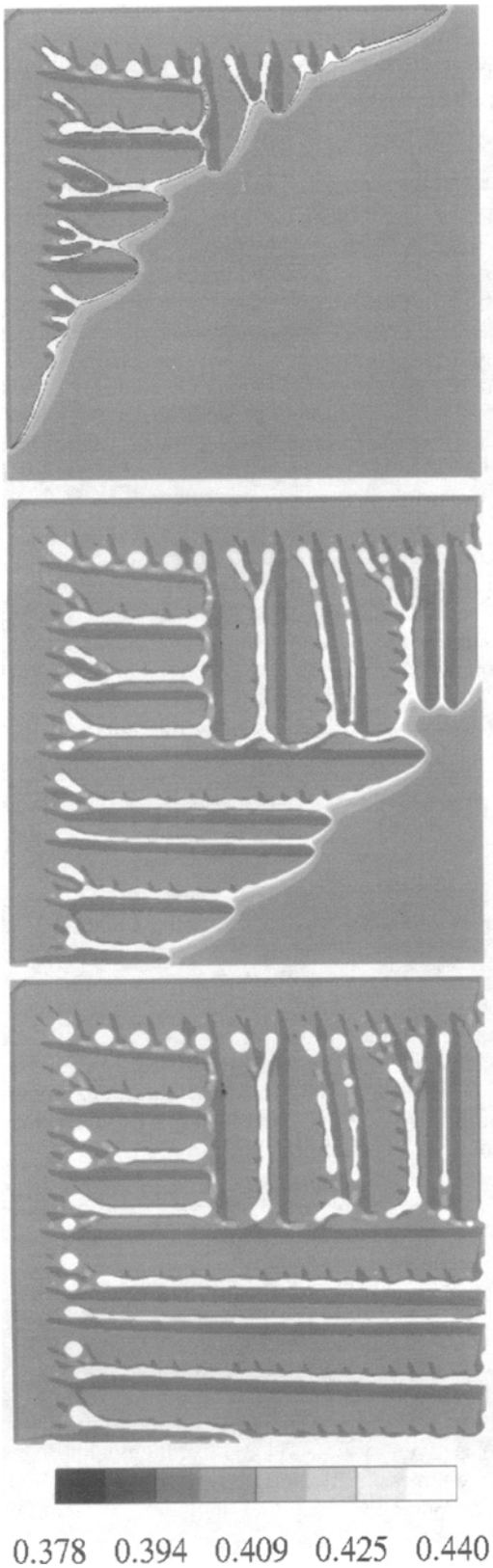


Fig. 4—Plots of the concentration profiles for a nickel-copper alloy dendrite growing into a supersaturated melt at a fixed temperature shown at three successive times of 1.7, 2.8, and 5.1 ms. The calculation was done on a 500 × 500 grid and corresponds to a box 22.9 μm on a side. $D_S/D_L = 10^{-4}$.

$$\frac{\partial f}{\partial c} = \text{constant} \qquad [23]$$

For a selected temperature, simultaneous solution of Eqs. [11] and [23] in one dimension show that ϕ will vary from 0 to 1 across a stationary interface between liquid and solid as c varies between two specific compositions c_L and c_S. These two compositions have been shown[1] to agree with those obtained by classical thermodynamic methods applied to equilibrium between bulk phases and given by the common tangent construction to the liquid and solid free energies at the selected temperature. The common tangent line to $f(0,c,T)$ and $f(1,c,T)$ at fixed T is the projection of the tangent plane to the surface $f(\phi,c,T)$ as shown in Figure 3. More complex forms for $f(\phi,c,T)$ are employed to treat eutectic solidification.[7–10]

For time-dependent solutions, the simplest postulate which guarantees that global equilibrium is achieved at long times, that F decreases monotonically with time, and that solute is conserved is

$$\frac{\partial c}{\partial t} = \nabla \cdot [M_c c(1 - c)\nabla(\frac{\partial f}{\partial c})] \qquad [24]$$

where M_c is positive and may depend on T and ϕ. If ϕ is identically 0 or 1 (single-phase liquid or solid). Eq. [24] is the ordinary solute diffusion equation

$$\frac{\partial c}{\partial t} = D\nabla^2 c \qquad [25]$$

assuming $D = M_c \mathrm{R}'T$ is constant. To permit different solute diffusivities in the liquid and solid, Warren and Boettinger[4] have used

$$M_c = \{(D_S + p(\phi)(D_L - D_S)\}/\mathrm{R}'T \qquad [26]$$

where D_L and D_S are the diffusion coefficients in the bulk liquid and solid phases and $p(\phi)$ is a smoothed step-function such that $p(0) = 1$ and $p(1) = 0$.

One can add a gradient energy term for the concentration in addition to the gradient term for phase field into Eq. [22]. This addition was explored by Wheeler et al.[2,3] in the context of solute trapping at a planar interface for high-velocity solidification. Behavior similar to the Aziz model[42,43] for solute trapping was obtained. Langer and Sekerka[44] employed a solute gradient energy term (and no phase field) in a model for a moving solid-solid interface in a miscibility gap system. For a single-phase alloy, inclusion of this term leads to the fourth-order diffusion equation or Cahn–Hilliard equation used for spinodal decomposition or diffusion in thin multilayers (with different forms for the free-energy function, f).[45,46]

IV. RESULTS FOR ISOTHERMAL ALLOY DENDRITIC GROWTH

For comparison to results presented in the next section, we show previous results[4] of solving equations similar to Eqs. [15] and [24] for dendritic growth into a supersaturated liquid at a prescribed temperature (Eqs. [3.2] and [3.8] of Reference 4). Thus, there is no heat flow or recalescence during dendritic growth in this case. As a result of the finite size of the computational domain, dendritic growth slows when the solute fields in front of the dendrite tips interfere

with the domain boundary where a no-flux solute condition is applied. Thus, for simulation of free dendritic growth, assessments of tip radius and velocity must be made before this interference. On the other hand, we can take advantage of the collision of the dendrite with the boundary of the computational domain as it mimics the situation that occurs during equiaxed dendritic growth in castings.[48] The domain boundary may be viewed as a surface midway between adjacent nuclei from which dendrites grow and subsequently impinge.

Alloy calculations were performed using melting points, latent heats, and surface energies appropriate to Ni and Cu for components A and B, respectively.[1] The concentration variable is taken to describe the atomic fraction of Cu. The phase diagram of the real alloy is lens shaped with solution behavior that is nearly ideal, so our simple alloy free energy is a reasonable approximation. The value of μ was taken as ~0.4 cm/sK, which is appropriate for interface kinetics governed by diffusive jumps, but which is a factor of 50 smaller than the preferred value based on collision-limited kinetics. This choice was necessary to limit computation times. The interface thickness, δ, was taken as 30 nm. The ratio D_S/D_L was taken as 10^{-4} appropriate to a substitutional solid solution. The strength of the noise ($\alpha = 0.3$) and the anisotropy ($\gamma_e = 0.04$) were taken as described in Reference 4. A more complete discussion of the computational limitations can also be found in Reference 4 as well as results for $D_S/D_L = 10^{-1}$, which correspond to a hypothetical alloy with interstitial solid diffusion.

Figure 4 shows the calculated growth shapes and the solute segregation patterns that develop at different times for an alloy with 0.40831 fraction Cu at a temperature of 1574 K and thus a dimensionless initial supersaturation of 0.86. Figure 5 shows the growth temperature and average alloy composition superimposed on the phase diagram. In order to permit the maximum distance for the development of a dendritic structure, the seed was placed in the corner of the computational domain rather than at the center. We employed a triangular seed region with $\phi = 1$ and with concentration equal to the liquid. The procedure of calculating only one quarter of the growth shape is possible because the boundary condition employed for the phase field (normal derivative equals zero) enforces a symmetry condition at the walls that preserves the fourfold anisotropy. For presentation the calculated growth shape can be reflected twice across the box boundaries to obtain a complete dendritic morphology. Growth patterns started at the center of a computational domain are similar but do not contain the synthetic twofold symmetry axes regarding the details of the side-branch structure. Because the temperature of the system is fixed, the fraction of solid cannot exceed 0.86 at the end of solidification.

The calculated patterns show many features similar to those observed in dendritic growth experiments using transparent materials and in micrographs of cast alloys. The central spines of the primary stalks and the fully developed secondary arms have the lowest concentration. The solid concentration generally increases away from these spines out toward the remaining liquid. However, in the vicinity of secondary arms, the simulations show that the concentration does not rise monotonically with distance along lines drawn from the center of the primary stalk into the spine of the secondary arm. We note the tendency of the side branches to begin their growth in directions not exactly perpendicular to the primary stalk, and to later develop growth axes that are perpendicular to the primary stalk. The competition between adjacent secondary arms leads to the coarsening of the secondary arm spacing away from the tip region. The roots of the surviving secondary arms near the primary stalk are pinched in a manner similar to that commonly observed in real dendrites. The side arms are also asymmetric with respect to this central spine with the region of the secondary arm toward the front of the dendrite being thicker. We also note the coalescence of adjacent and impinging secondary and tertiary arms. This coalescence process occurs in dendritic structure where little or no eutectic forms, as is the case for the lens-type phase diagram used here. It should be noted that the occurrence of coalescence is expected to be very sensitive to the value of δ used for the calculations.

Figure 5 shows histograms of the fraction of the computational domain contained in a set of concentration bins for the patterns shown in Figure 4, top and bottom, respectively. The relevant portion of the phase diagram is also shown at the top of Figure 5 along with the initial concentration and the (fixed) growth temperature. The histogram for Figure 4 (top) contains three major peaks with increasing levels of concentration that correspond to solid, extra-

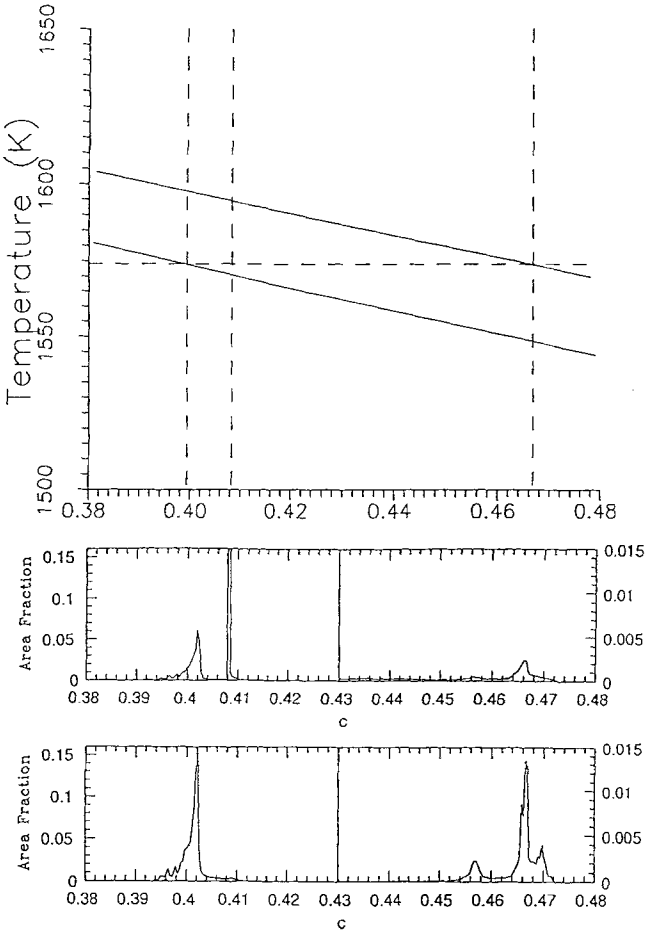


Fig. 5—Relevant portion of the phase diagram (top) and concentration histograms (center and bottom) for Fig. 4 top and bottom. An expanded vertical scale is used to the right of the vertical line in the center of the histograms. The equilibrium concentrations for the solid and liquid at a stationary flat interface at 1574 K are 0.3994 and 0.4668, respectively. The initial concentration was taken as 0.4083.

dendritic liquid, and interdendritic liquid, respectively. The solid peak is relatively broad, containing information regarding the microsegregation pattern in the frozen solid. It contains concentrations below and above the equilibrium solid composition for a stationary, flat interface. The concentration in the solid at the dendrite tip is the lowest observed in the solid. Its value is the combined effect of the shift in the solid composition due to curvature and kinetics. The central peak is quite sharp and corresponds to the bulk of the liquid that still remains at the original alloy concentration. We note a slight tail on the high-concentration side corresponding to the buildup of solute around the dendrite envelope. The peak at the highest concentration corresponds to the interdendritic liquid between the dendrite side branches. In the histogram for Figure 4 (bottom), the central peak is gone. The peak at low solute concentration corresponds to the solid. The interdendritic liquid peak contains a small spread in composition reflecting the positive and negative curvatures of the interface between the liquid and solid.

V. RECALESCENCE CALCULATIONS

In this section, we explore the use of the phase-field equations to describe alloy dendritic solidification when the temperature is not held fixed. To solve for evolution of the temperature distribution using Eq. [4] simultaneously with the equations for the ϕ and c is not practical for realistic values of the thermal diffusivity, given the current numerical approach and computational resources. Thus, we consider the situation where the temperature is assumed to remain spatially uniform at each instant and where heat is removed at a constant rate from the computational domain. The heat equation is then replaced by a simple statement of heat balance that involves the heat extraction rate and the rate at which the fraction of solid increases. This approximation has some justification in the fact that the diffusivity of heat is 10^3 to 10^4 larger than for the solute diffusivity. However, at the high rates of growth associated with large undercoolings employed here, local recalescence at the dendrite tip cannot in reality be neglected. Thus, we view the present approach as a first improvement of the isothermal model for alloys.

We consider a liquid that is being cooled at a rate, $\dot{T}$, prior to the introduction of a seed at $t = 0$ at a temperature, T_N. At subsequent times during solidification, the temperature-time history is a balance between the imposed external heat extraction rate, defined by $\dot{T}$, and the rate of release of the latent heat. For the computations, the control parameters are $\dot{T}$, T_N, and the size of the computational box.

Such a scheme is quite similar to the assumptions made in various microscopic models of equiaxed solidification that are coupled to macroscopic heat transfer codes.[48] For each node the nucleation temperature is generated by a nucleation model and the heat extraction rate is set by the heat flow to/from adjacent nodes of the macroscopic heat-flow code. The nucleation model also determines the number of nuclei/unit volume and thus sets the volume in which a single grain grows. This volume may be associated with the size of the computational domain used for the present calculations.

For our calculations, the thermal balance for the domain is given by

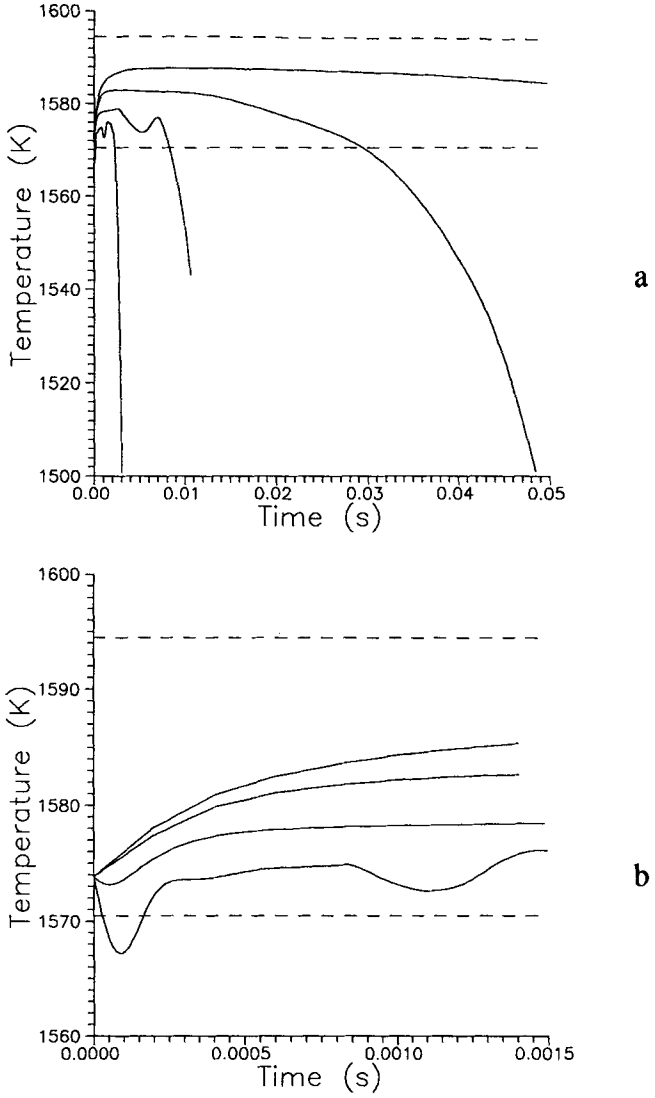


Fig. 6—$T(t)$ behavior during solidification for a computational domain of 11.4 μm on a side and a value of T_N = 1574 K. Results for four different values of the cooling rate, $\dot{T}$, 2.1 × 10^3, 8.4 × 10^3, 3.4 × 10^4, and 1.3 × 10^5 K/s are shown from top to bottom. The general trend of recalescence and subsequent cooling is noted in all four cases. An expanded view of the early time behavior is also shown.

$$\frac{dT}{dt} = -\dot{T} + \frac{L}{C_p}\frac{d}{dt}\left[\frac{1}{A}\int_A \phi(\bar{x},t)\,dA\right] \qquad [27]$$

where the expression in square brackets is the fraction solid, f_s, A is the area of the computational box (for this two-dimensional calculation), L is the latent heat of fusion, and C_p is the heat capacity per unit volume. This equation was evaluated simultaneously with equations similar to Eqs. [15] and [24] given in Reference 4. We use the same materials data for the Ni-Cu alloy employed for the isothermal simulations. The alloy concentration is again taken to be 0.40831 fraction Cu. The computed liquidus and solidus for this composition are 1594.5 and 1570.5 K.

Figure 6 shows full- and short-term $T(t)$ behavior obtained during growth in a computation domain 11.4 μm on a side, using a value of T_N = 1574 K, $D_S/D_L = 10^{-4}$, and for four different values of the cooling rate. The selected values of the initial supercooling and cooling rate are quite large in order to facilitate numerical computations. The

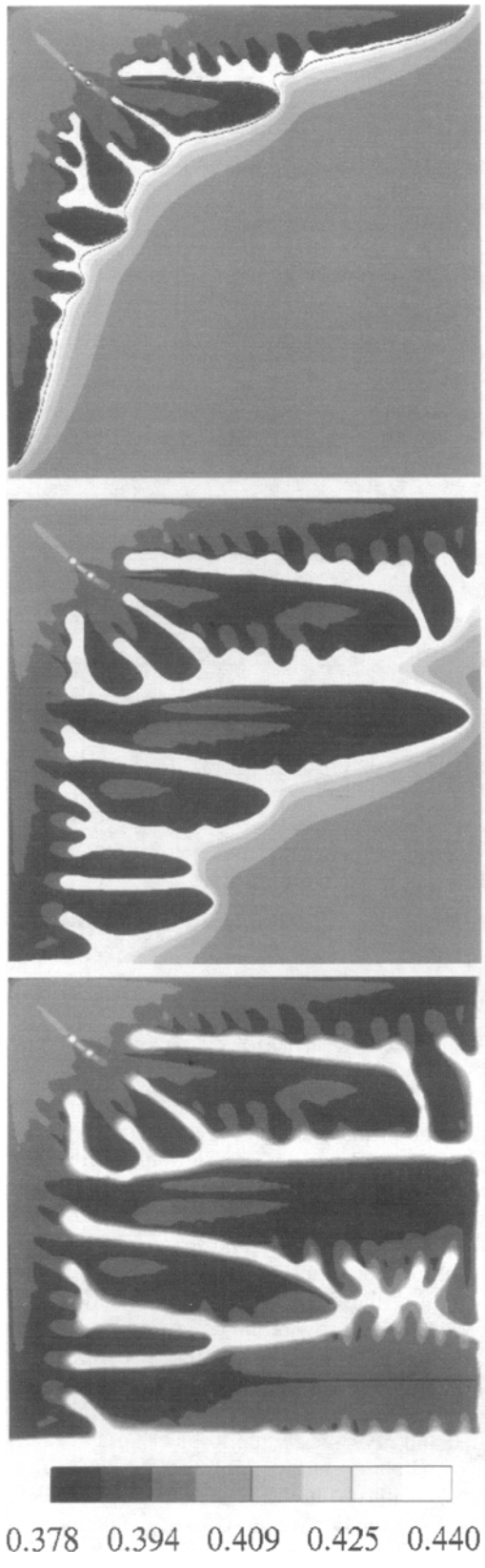


Fig. 7—Growth morphology and solute patterns for $\dot{T} = 3.4 \times 10^4$ K/s at times of 2.5, 5.6, and 12.1 ms. The last corresponds to the time when the fraction solid reaches unity. $D_S/D_L = 10^{-4}$.

general trend of recalescence and subsequent cooling is noted but the simulated growth morphologies vary considerably among the four cooling rates. At the lowest cooling rate, the growth is globulitic with only slight instability of the growing shape into cell-like fingers but with no secondary side-arm formation. As the cooling rate is increased, the growth shape exhibits increased instability and, at the highest cooling rate shown, the growth shape resembles a well-developed dendritic structure. Collisions of the primary stalks and side arms with the walls of the computational domain cause momentary delays in the rate of increase of solid fraction that lead to the multiple recalescence humps for the two higher values of the cooling rate. These multiple recalescence humps are more common when the computational domain is small. The early time behavior shows that for the two highest cooling rates, the temperature continues to fall before the latent heat release rate is sufficient to cause recalescence. In the other cases, the temperature-time curves begin with a positive slope because the initial growth rate of the seed is sufficiently fast to overpower the imposed heat extraction rate.

Figures 7 and 8 show snapshots of the growth morphology and solute patterns for two ratios of D_S/D_L of 10^{-4} and 10^{-1} and for $\dot{T} = 3.4 \times 10^4$ K/s in a computational box 22.8 μm on a side. The temperatures and times corresponding to the three snapshots are as follows: in Figure 7, 1577.0 K, 2.5 ms, 1576.9 K, 5.6 ms, and 1499 K, 12.1 ms; and in Figure 8, 1576.7 K, 2.5 ms; 1576.2 K, 5.6 ms, and 1531.0 K, 11.2 ms. The last snapshot for each corresponds to the time when the fraction solid reaches unity. As a result of the increased level of solid diffusion employed for Figure 8, solidification is completed at a higher temperature, which would be expected. In order to depict the concentration profile in black and white, we have chosen to employ a staircase functional relationship between concentration and the gray scales. We have also chosen to place all concentrations above 0.44 in the last gray scale so as to depict the smaller concentration gradient in the liquid.

Figure 9 shows the $T(t)$, $f_s(t)$, and $T(f_s)$ plots obtained for the calculations shown in Figures 7 and 8. Also shown for reference is the $T(f_s)$ relation calculated from a Scheil model integrated numerically for the lens-shaped phase diagram. The rising portions of the $T(t)$ curves up to about 2.5 ms correspond to the growth of the primary stalks out to the domain boundary (Figures 7 and 8, top). The formation of a large amount of solid during the relatively flat portion (plateau) of the $T(t)$ curves corresponds to the filling in of the secondary and tertiary arms (Figures 7 and 8, center). Apparently, a balance between the growth rate of the tips and the external heat extraction rate is found by the system to maintain a nearly constant temperature. The temperature finally begins to drop only after the fast growth mode of the secondary and tertiary arms is halted by collision with the walls of the computational domain. During this final period, the side arms thicken, completing solidification. Interestingly, the fraction solid *vs* time curve is nearly linear over a large variation in the growth pattern in a manner similar to that obtained by Rappaz and Thevoz.[47]

The results show that the spacings of surviving secondary arms are not as small as was obtained for isothermal growth. This may be due to the recalescence effect, which reduces the undercooling compared to the initial value and hence reduces the source of instability. Also in contrast to the isothermal results, one notes that the concentration decreases down the length of the primary dendrite stalks, indicative of the increasing temperature of the tip during

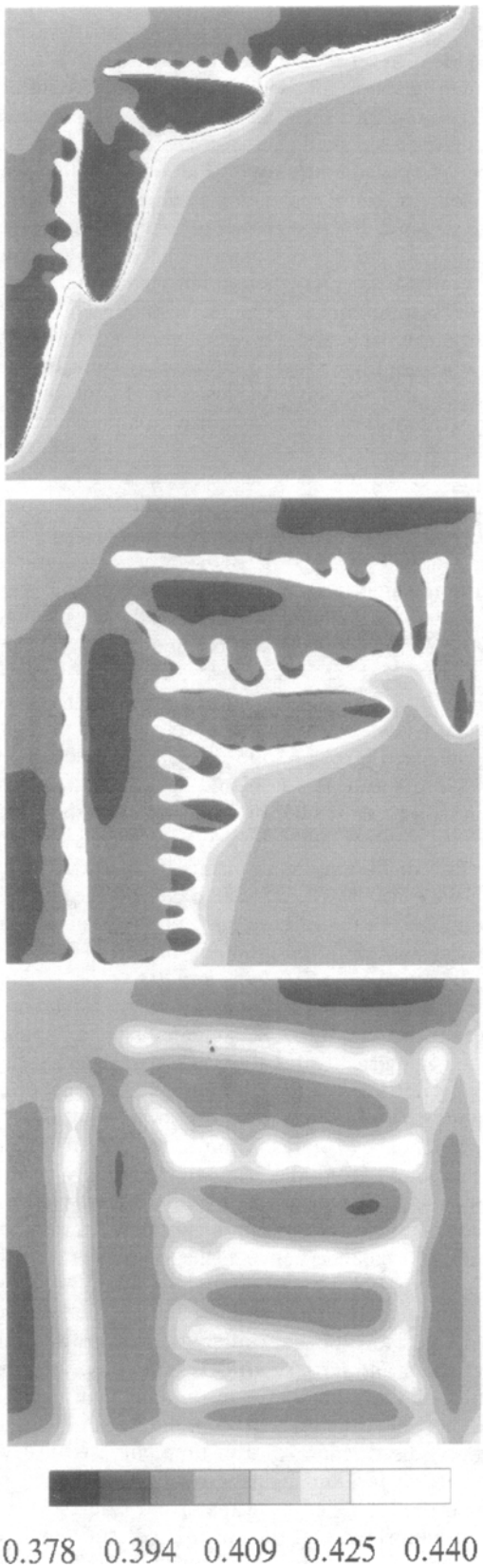
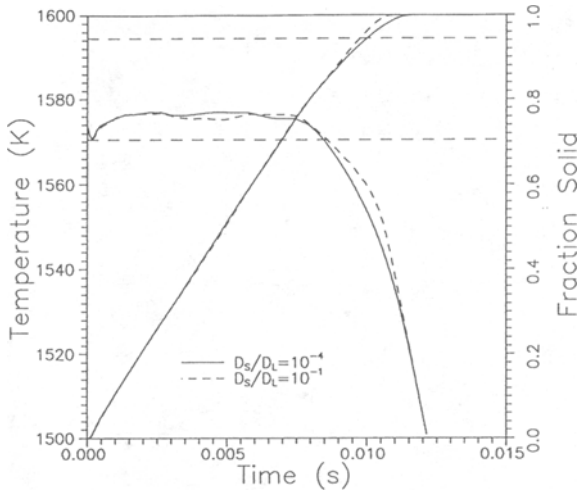
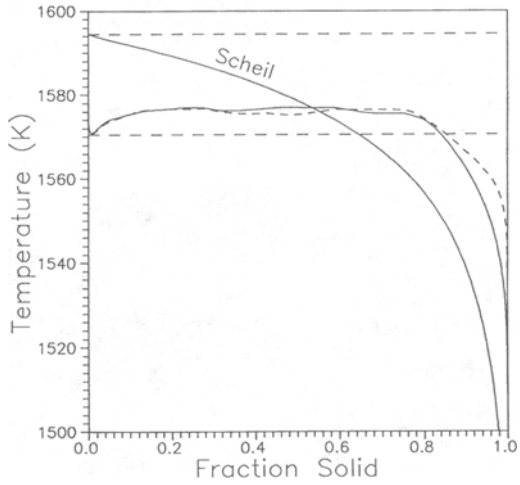


Fig. 8—Growth morphology and solute patterns for $\dot{T} = 3.4 \times 10^4$ K/s at times of 2.5, 5.6, and 11.2 ms. The last corresponding to the time when the fraction solid first reaches unity. $D_S/D_L = 10^{-1}$.



a



b

Fig. 9—(*a*) Temperature-time and fraction solid-time, (*b*) temperature-fraction solid curves for the two cases in Figs. 7 and 8 along with a calculation of the temperature-fraction solid curve from a Scheil model of solidification (integrated numerically for the lens-shaped phase diagram).

growth of the primary stalks. Figure 8 shows morphologies similar to Figure 7 but with a more uniform concentration profile remaining in the solid because of the increased level of solid diffusion.

Also shown in Figures 7 and 8 (top) are curves for the isoconcentrate corresponding approximately to the liquid composition at the dendrite tip (0.44). In many models of equiaxed dendritic growth, the composition of the interdendritic liquid is assumed to be completely mixed inside this isoconcentrate. Figure 10(a) shows an enlargement from the interdendritic region of Figure 7 top (at 2.5 ms) using an expanded and reversed (white to black) gray scale. The rejection of solute by the dendrite causes the concentration in the extradendritic liquid to rise from the bulk value of 0.408 far from the dendrite to 0.44 at the dendrite envelope defined by the tip concentration. The concentration in the interdendritic liquid rises from 0.44 to 0.464. The steepest gradient occurs over a distance of approximately two dendritic tip radii behind the dendrite envelope boundary. The remaining variations in the interdendritic liquid concentration extend much further back from the envelope and are attributable to the variations in curvature that drive the coarsening processes of the side arms. Note that the regions

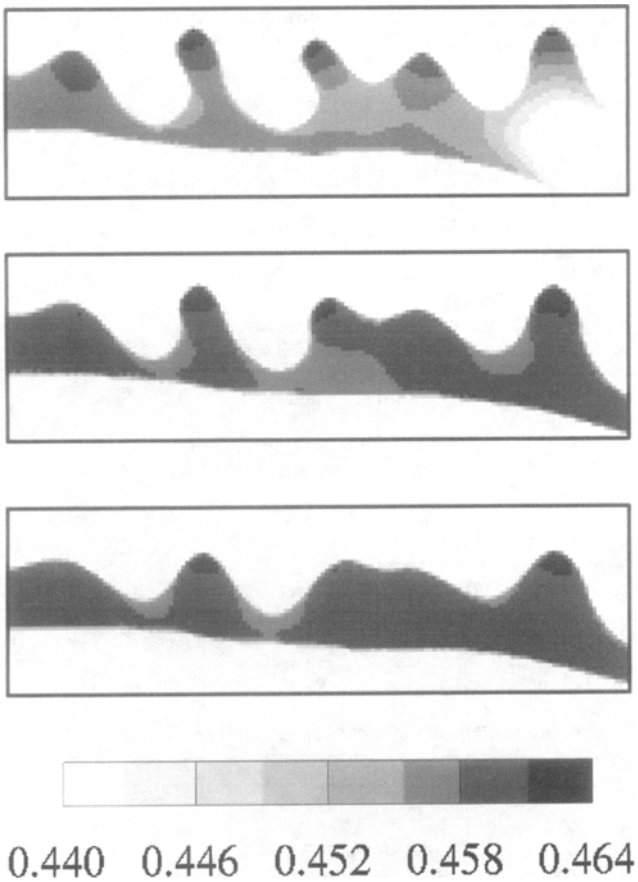


Fig. 10—Enlargements from the interdendritic region at (*a*) 2.5 ms (Fig. 7(*a*)), (*b*) 3.1 ms, and (*c*) 3.7 ms showing the small variations of concentration in the liquid and the disappearance of a secondary arm.

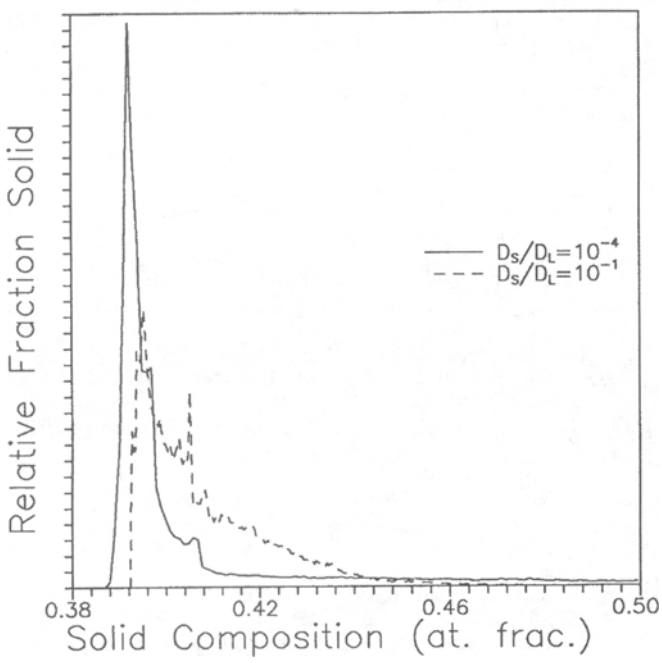


Fig. 11—Histograms of the solid composition at the time when the fraction solid first reaches unity for the case of the two diffusion ratios (Figs. 7 and 8 bottom).

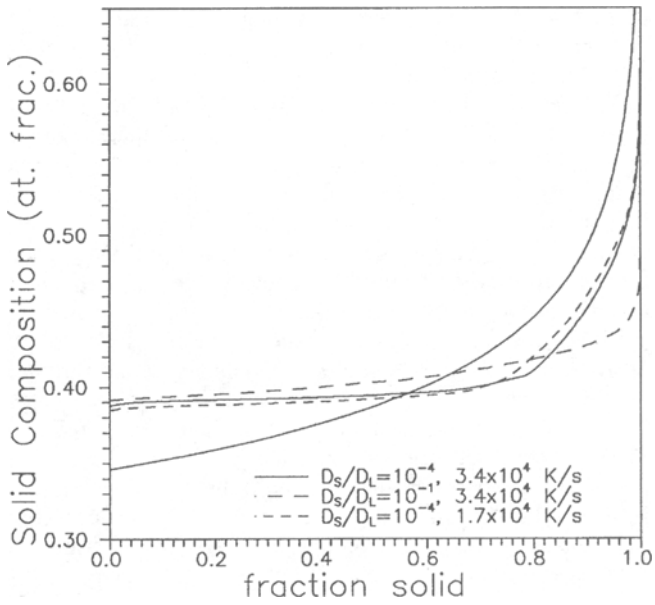


Fig. 12—Integrated histograms showing the fraction of solid with a composition less than or equal to the values indicated on the curves. Four curves are shown. Two correspond to Figs. 7 and 8 bottom ($\dot{T} = 3.4 \times 10^4$ K/s) and another corresponds to a calculation done for a smaller cooling rate of 1.7×10^4 K/s. The fourth curve shows the prediction from the Scheil model.

where the interface is concave toward the liquid have the highest concentrations as expected from the Gibbs–Thomson effect. It should be remembered that this dendrite is particularly small compared to those in ordinary castings and thus the concentration variations in the interdendritic region due to curvature are consequently larger.

Figures 10(b) and 10(c) are enlargements of the same region at later times (3.1 and 3.7 ms). During these times, the increase in fraction solid is occurring primarily at the dendrite tips and the temperature is approximately constant. Thus, the amount of liquid in these enlargements is approximately constant. The sequence of enlargements shows the reduction of interface curvature and concentration gradients. In particular, the shrinkage of a side arm is shown.

One can be more quantitative regarding the microsegregation produced during the dendritic growth. Figure 11 shows histograms of the solid concentration obtained from all points in the computational domain at the time when the fraction solid reaches unity for the case of the two diffusion ratios. The histograms were truncated at a solid concentration of 0.5 for presentation (the tail for $D_S/D_L = 10^{-4}$ persists out to approximately 0.63). One notes a much more peaked distribution for the case of limited solid diffusion but a much longer tail at high concentration (not shown). We believe that the small peaks in the histograms near the initial liquid composition (0.408) are due to solute trapping during the early-stage growth of the seed crystal.

An integrated form of the two histograms is shown in Figure 12 along with an addition curve for a growth simulation for a smaller value of $\dot{T} = 1.7 \times 10^4$ K/s. These curves give the fraction of solid with a concentration less than or equal to the indicated values. Plots of solid concentration *vs* fraction solid are widely used to describe microsegregation in the solidification field and the simple Scheil calculation for the lens-type phase diagram is shown in Figure 12 for comparison. It should be noted, however,

that the solid concentration at the interface may not increase monotonically with time as is often implied in such plots generated by simple microsegregation models.

The concentration at zero fraction solid in Figure 12 represents the concentration near the core of the dendritic arms. The simulated values are significantly higher than is predicted from the Scheil calculation. This effect is well known and is due to the kinetics of the dendrite tip.[49] It can be seen that this concentration is only slightly increased by the orders of magnitude change in the level of solid diffusion at the high cooling rate of 3×10^4 K/s. At larger values of the fraction solid, one notes extremely flat solute profiles, as have been found experimentally[50] and through modeling for high rates of solidification.[49] One also notes that the difference between the two diffusion cases follows the trend expected from various back-diffusion models,[51,52] i.e., that the final solid to form is higher in concentration if the solid diffusion is slow. If the cooling rate is decreased by a factor of 2, the concentration at zero fraction solid is decreased slightly. This occurs because the amount of recalescence is increased (the trend shown in Figure 6) leading to a higher growth temperature, slower dendritic growth, and less deviation from the Scheil prediction. However, the factor of 2 reduction in the cooling rate only increases the plateau growth temperature by about 1 K, thus, the decrease in concentration at zero fraction solid is small.

An examination of the $T(f_s)$ curves over a broader range of cooling rates for the results in Figure 6 shows a strong approach to the Scheil prediction as the cooling rate is decreased.

VI. CONCLUSIONS

Compared to sharp interface models of solidification, the phase-field method employs an extra field variable to describe whether a specific location is liquid or solid. The burden of this additional variable and its associated equation is offset by the avoidance of the mathematically difficult free-boundary problem. Finite-difference calculations on a uniform mesh can then provide realistic-appearing simulations of dendritic growth.

For alloys, the solute pattern in the solid during and after solidification has been computed for dendritic growth that takes place in a system at fixed temperature and in a system subject to recalescence due to release of latent heat. These simulations were performed for relatively high initial supercoolings and cooling rates. Dendrites grown in a recalescing system under these conditions show variations in concentration down the primary dendrite stem. Based on a limited number of simulations, an increased secondary spacing is observed compared to dendrites grown isothermally and the recalescence curves exhibit a plateau temperature over which much of the dendritic growth occurs. Future improvements in the solution algorithm and/or the computation speed are required to fully reap the advantages of the phase-field method and to use it for smaller interface thickness, initial supercoolings, and imposed cooling rates.

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